IEOR E4004 Optimization Models and Methods

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Outline

- Post Optimality and Sensitivity Analysis
- Solvers and SA
- Applying Monte Carlo to LPs

Post Optimality and Sensitivity Analysis

Why Sensitivity Analysis?

- **Often the values of model parameters are just estimates from data**
- Sensitivity analysis investigates the impact changes in the parameters (a_{ij} , b_i, c_j) have on the optimal solution. There are three types of sensitivity analyses:
  - Identifying the sensitive parameters *(e.g., when even a very small change in the value of parameters has some impact on the optimal solution)*
  - Determining the range of parameters within which the optimal solution remains unchanged
  - Re-optimization *(e.g., finding the new optimal solution when there is a change in the parameters)*

Modified Problem

Modified problem is obtained by:

- changing the objective function,
- changing the right-hand side (rhs) of a constraint,
- adding a variable/activity, or
- adding a constraint.

Questions Of Interest

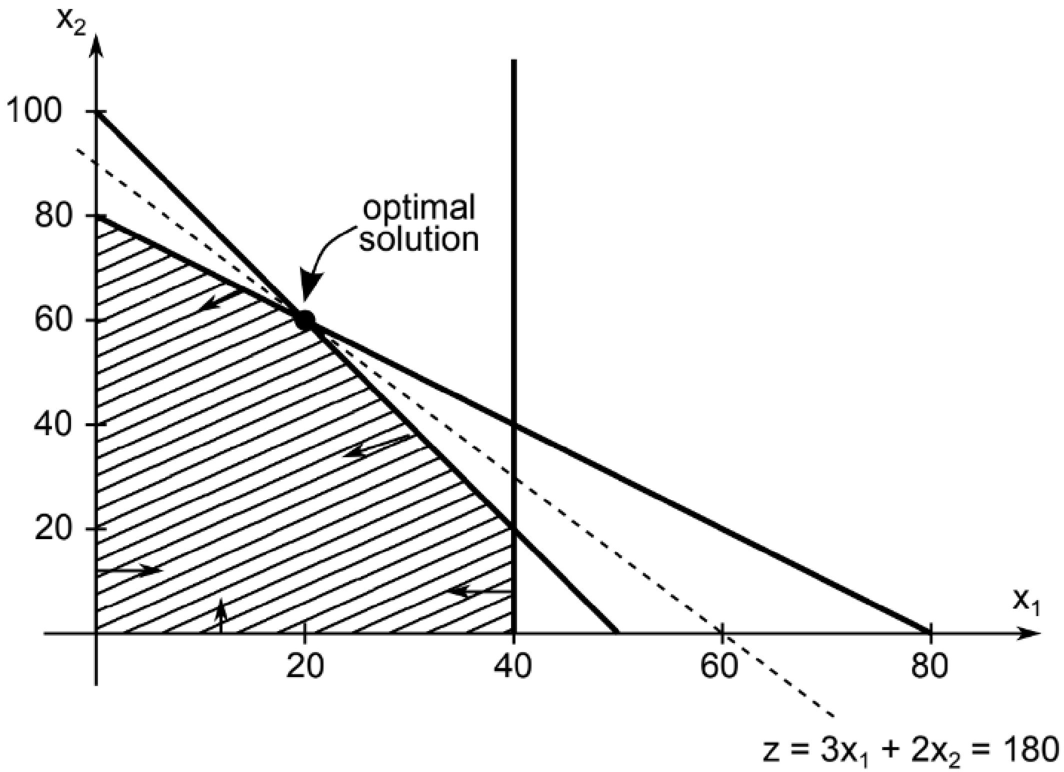
- For what changes is the original optimal solution also optimal in the modified problem?
- For what changes is it feasible?
- For what changes is the optimal basis also optimal in the modified problem?
- How do we recompute modified optimal solution from optimal basis?
- How do we recompute modified optimal tableau?
- If market conditions shift slightly, can we keep our current strategy or investment plan without change?
- Can we still operate under new business rules without breaking constraints like budgets, deadlines, or capacity?
- How do we update our decision model efficiently to reflect changes in pricing, resource limits, or goals?

Example: Sensitivity Analysis On Objective Coefficients

- Consider this LP:

$$\max z = 3x_1 + 2x_2$$

$$\begin{aligned} \text{s.t.} \quad 2x_1 + x_2 &\le 100 \\ x_1 + x_2 &\le 80 \\ x_1 \qquad &\le 40 \\ x_1, x_2 &\ge 0 \end{aligned}$$



Optimal solution at B: $x_1 = 20$, $x_2 = 60$, $z = 180$

Example: Sensitivity Analysis On Objective Coefficients

- Consider this LP:

$$\max z = 3x_1 + 2x_2$$

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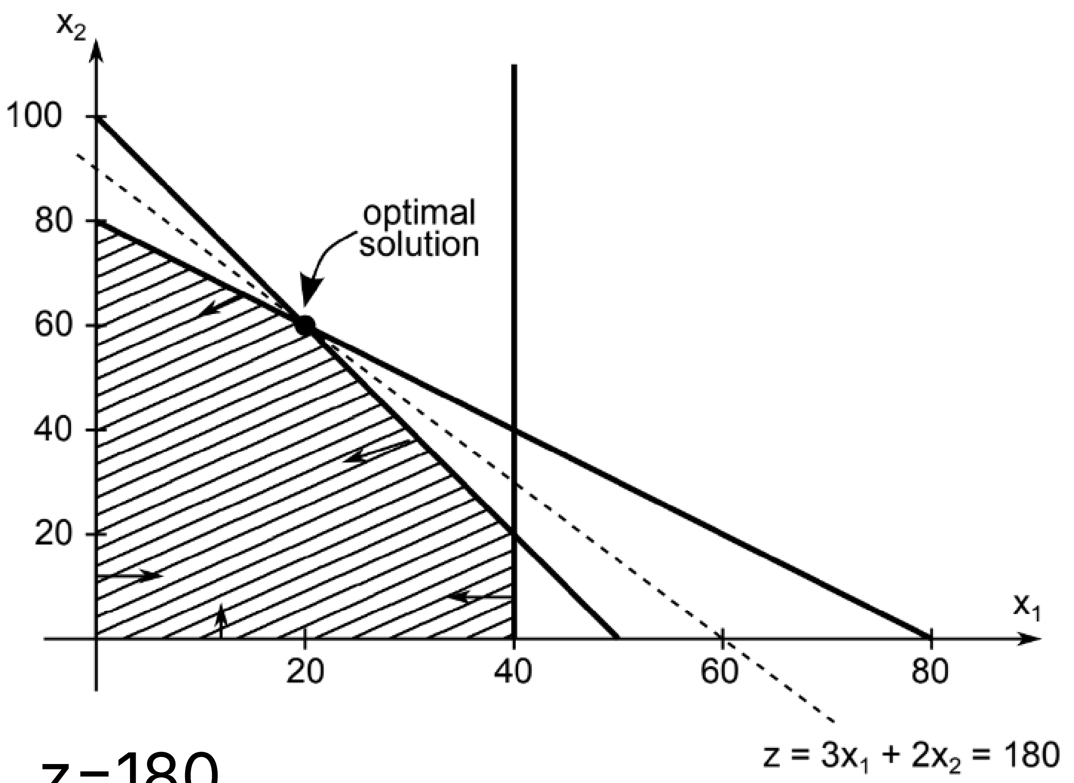
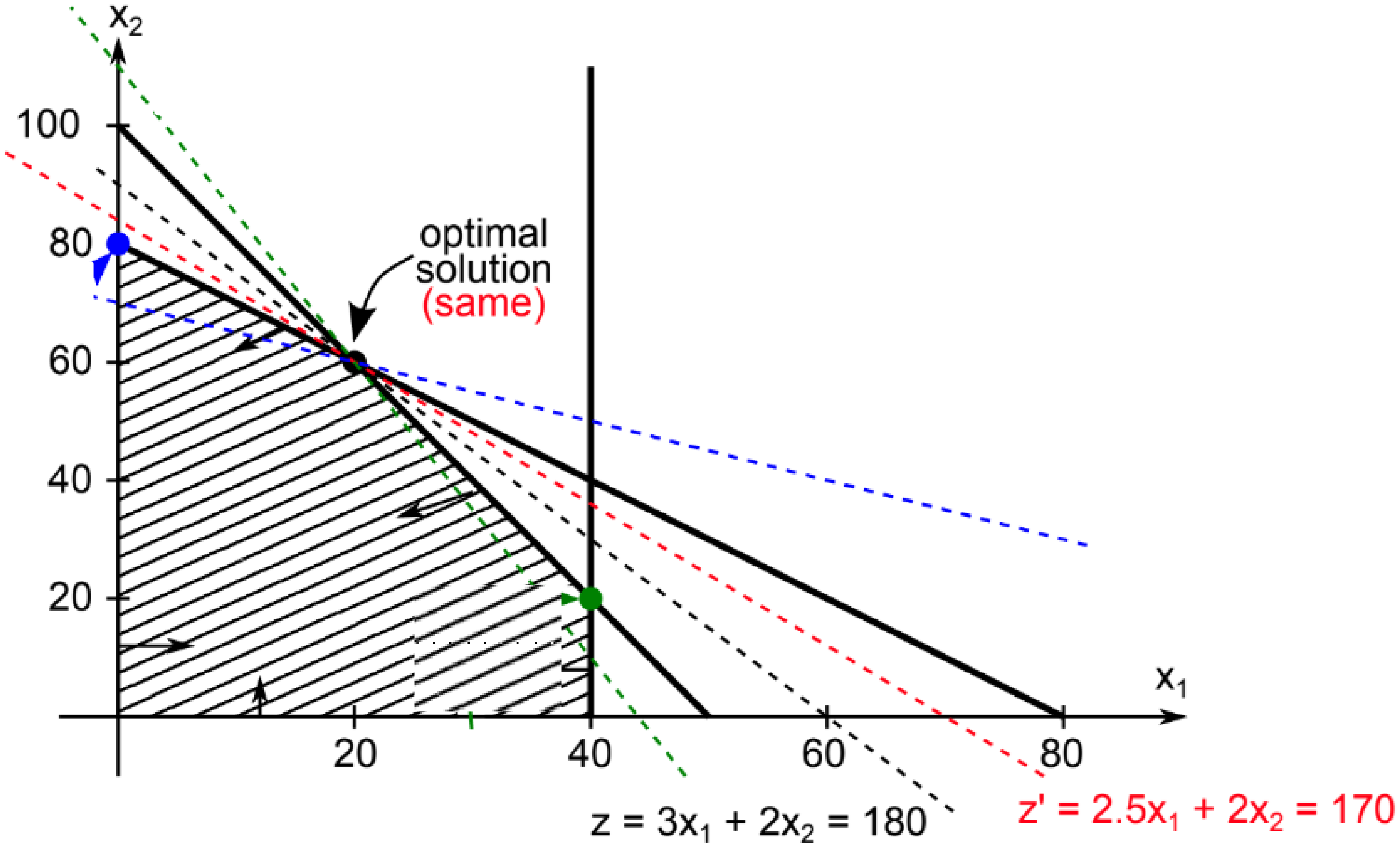


Optimal solution at B: $x_1 = 20$, $x_2 = 60$, $z = 180$

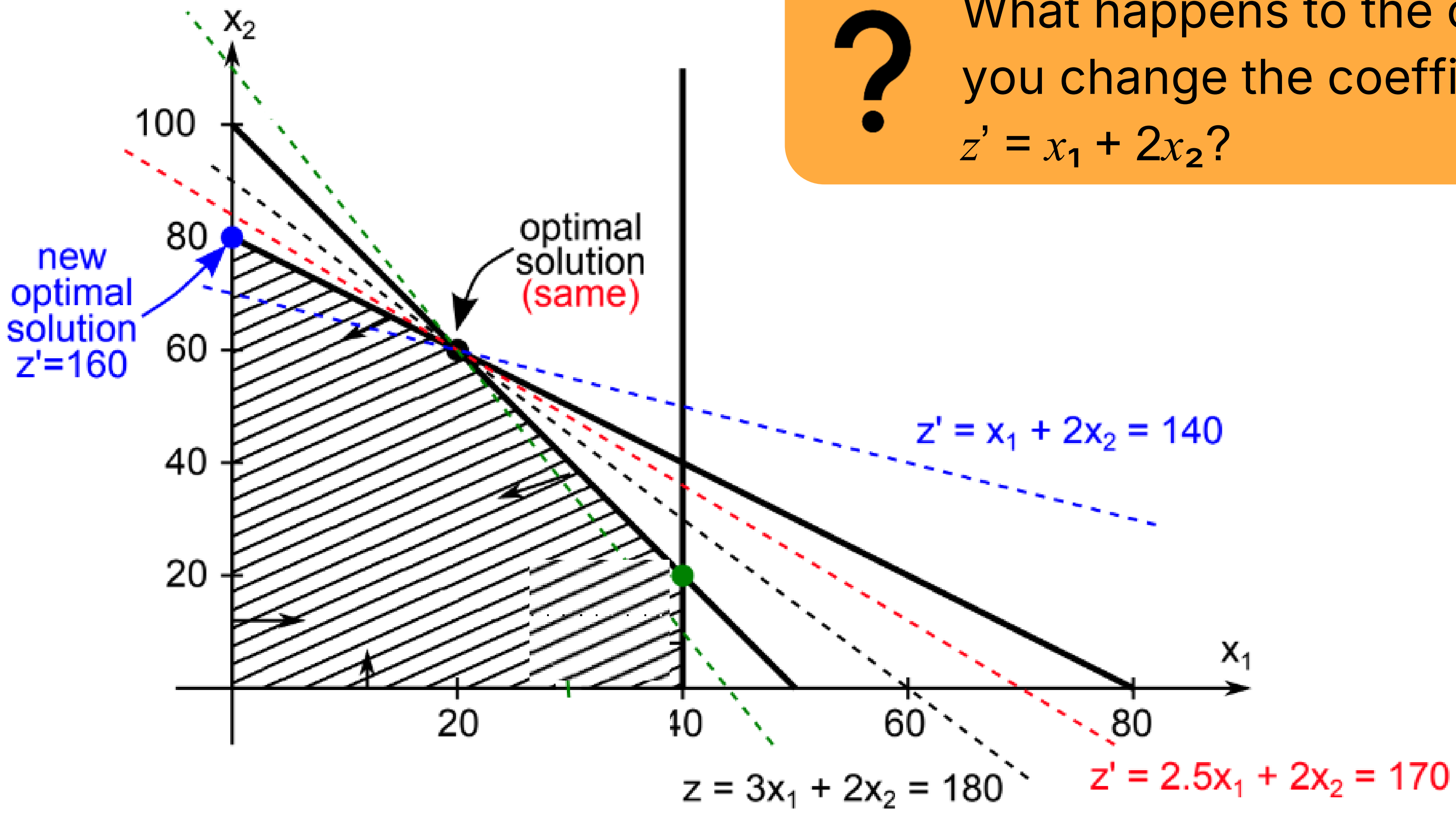
? What happens to the optimal solution when you change the coefficient $c_1 = 3$ in z to 2.5: $z' = 2.5x_1 + 2x_2$?

Example: Sensitivity Analysis On Objective Coefficients

The solution $x_1 = 20$, $x_2 = 60$ *still optimal, value* $z' = 170$



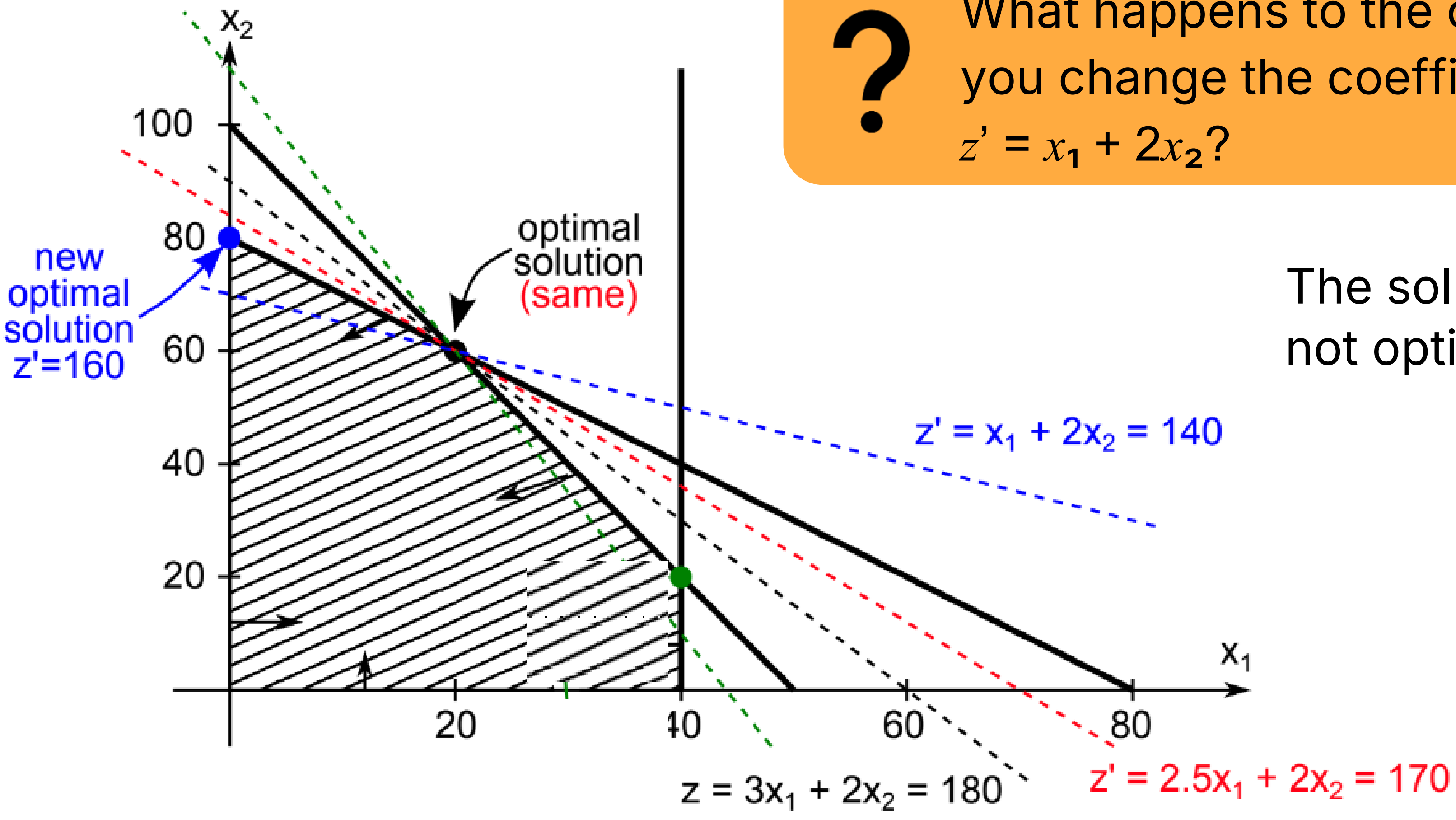
Example: Sensitivity Analysis On Objective Coefficients



?

What happens to the optimal solution when you change the coefficient c_1 = 3 to 1:
$z' = x_1 + 2x_2$?

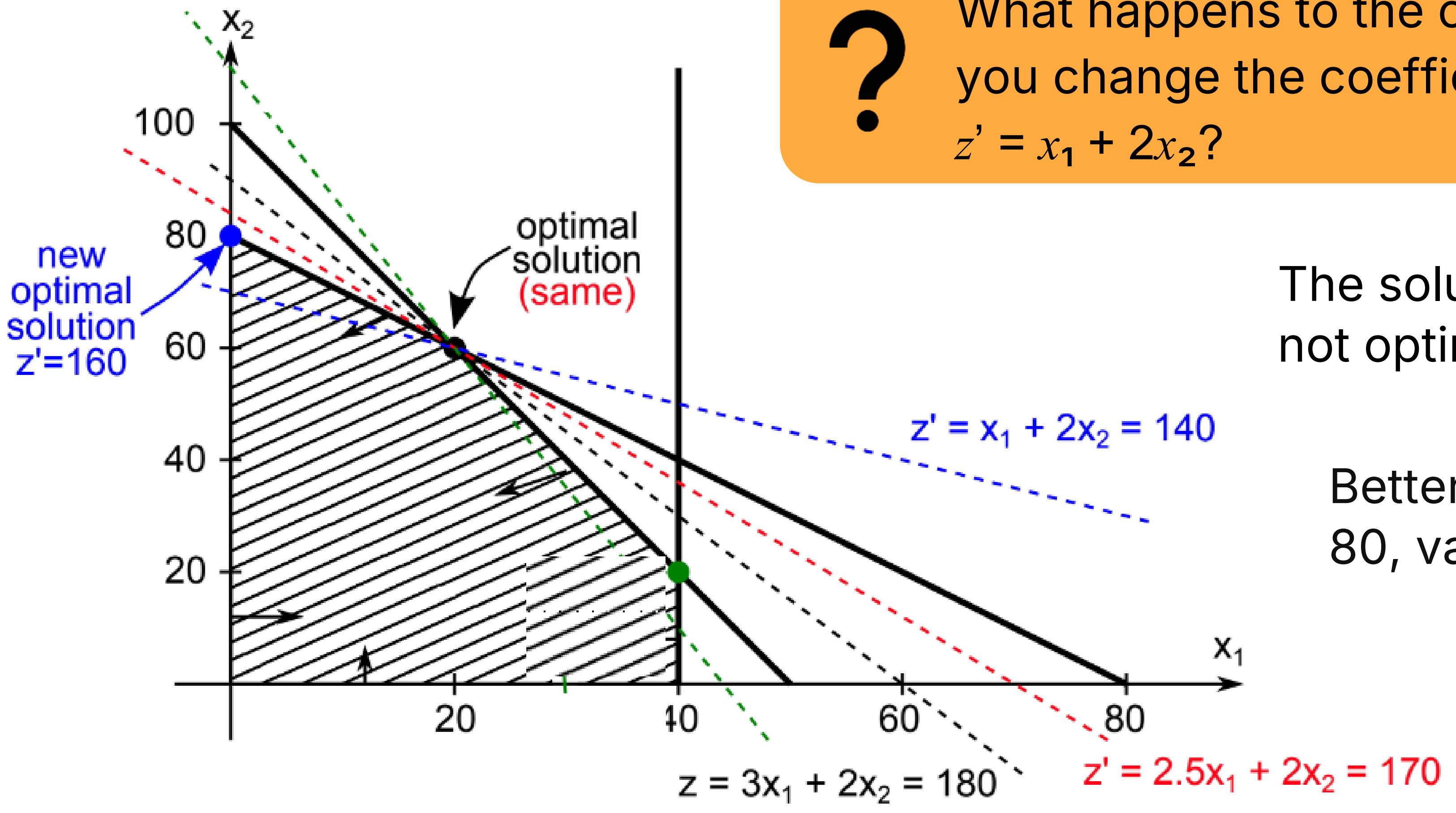
Example: Sensitivity Analysis On Objective Coefficients



? What happens to the optimal solution when you change the coefficient $c_1 = 3$ to 1: $z' = x_1 + 2x_2$?

The solution $x_1 = 20$, $x_2 = 60$ not optimal, value $z' = 140$

Example: Sensitivity Analysis On Objective Coefficients

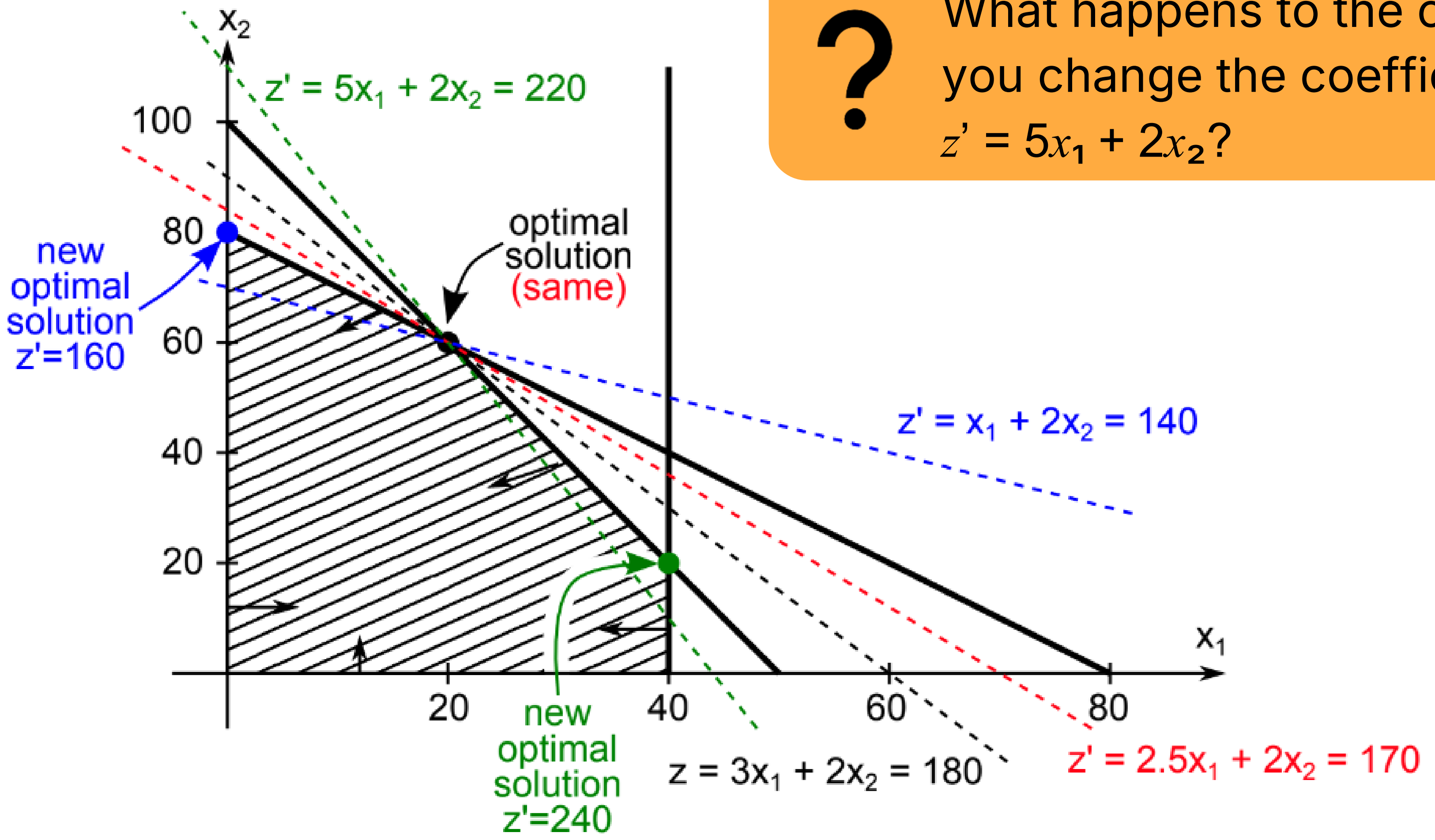


? What happens to the optimal solution when you change the coefficient $c_1 = 3$ to 1: $z' = x_1 + 2x_2$?

The solution $x_1 = 20$, $x_2 = 60$ not optimal, value $z' = 140$

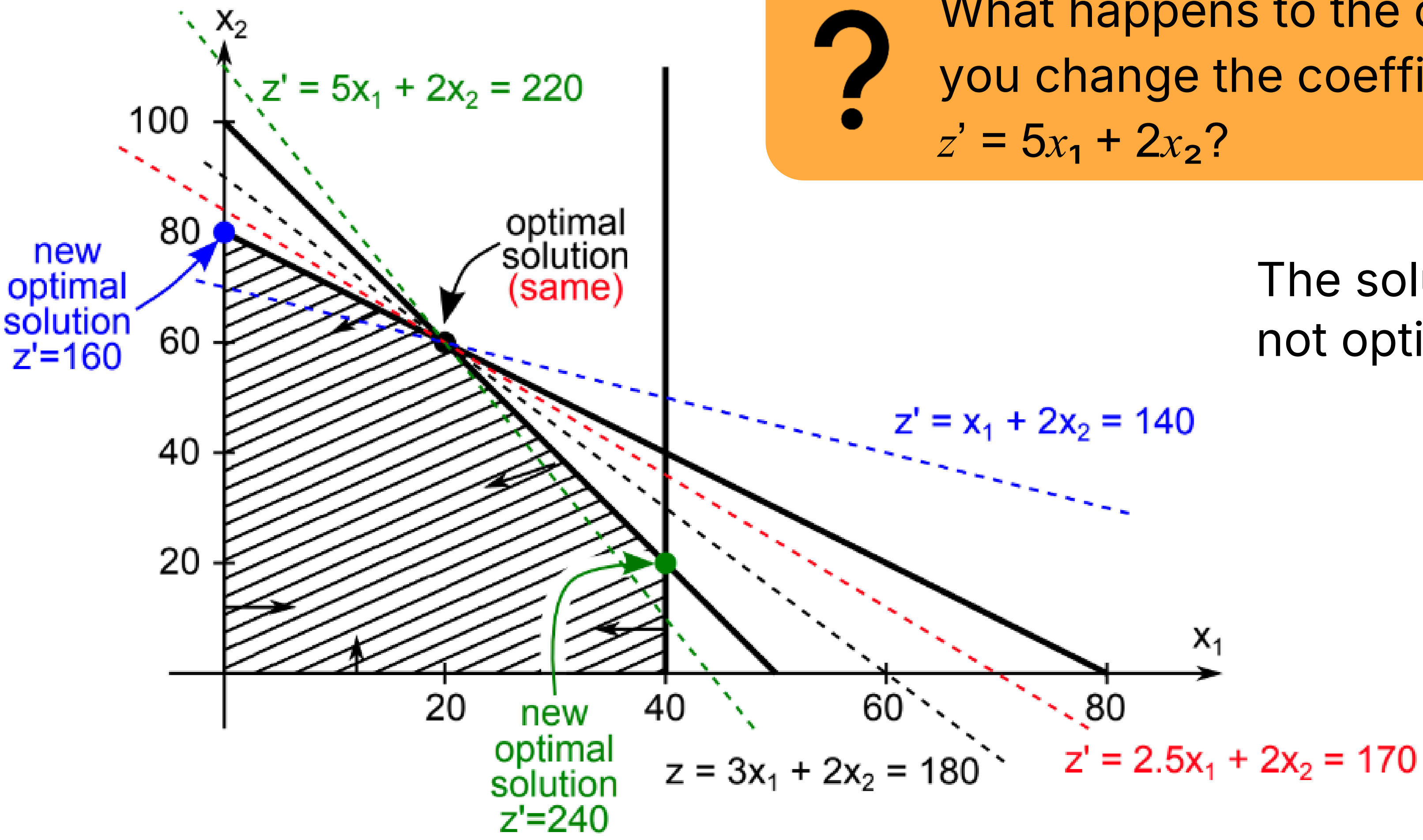
Better solution $x_1 = 0$, $x_2 = 80$, value z0 = 160 > 140

Example: Sensitivity Analysis On Objective Coefficients



? What happens to the optimal solution when you change the coefficient c_1 = 3 to 5: $z' = 5x_1 + 2x_2$?

Example: Sensitivity Analysis On Objective Coefficients

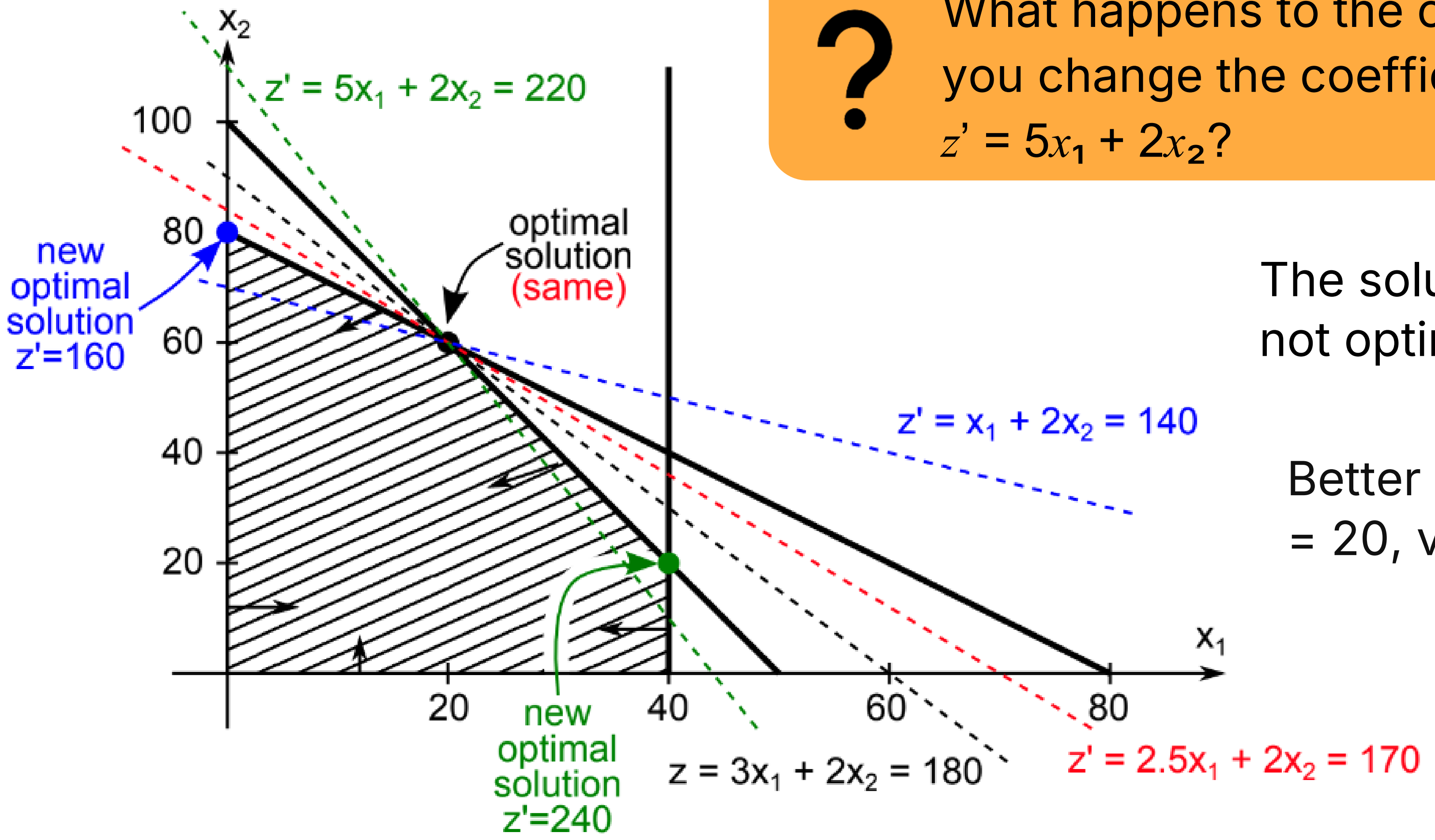


?

What happens to the optimal solution when you change the coefficient $c_1 = 3$ to 5:
$z' = 5x_1 + 2x_2$?

The solution $x_1 = 20, x_2 = 60$ not optimal, value $z' = 220$

Example: Sensitivity Analysis On Objective Coefficients



? What happens to the optimal solution when you change the coefficient c_1 = 3 to 5: $z' = 5x_1 + 2x_2$?

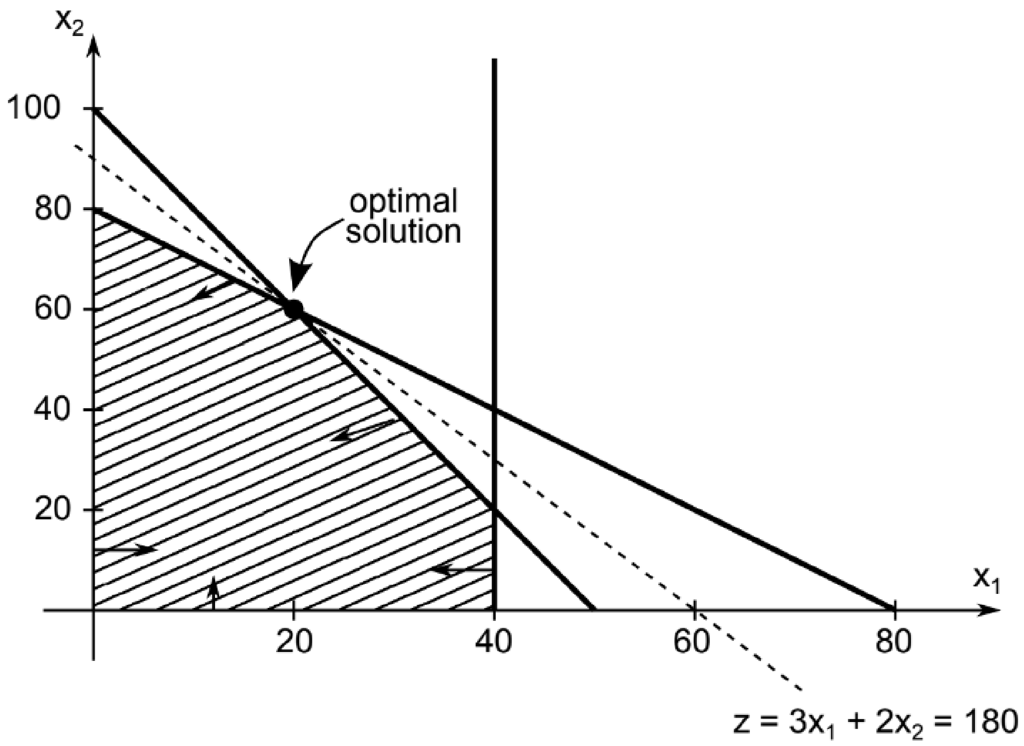
The solution $x_1 = 20$, $x_2 = 60$ not optimal, value $z' = 220$

Better solution $x_1 = 40$, $x_2 = 20$, value z0 = 240 > 220

Example: Sensitivity Analysis On Objective Coefficients

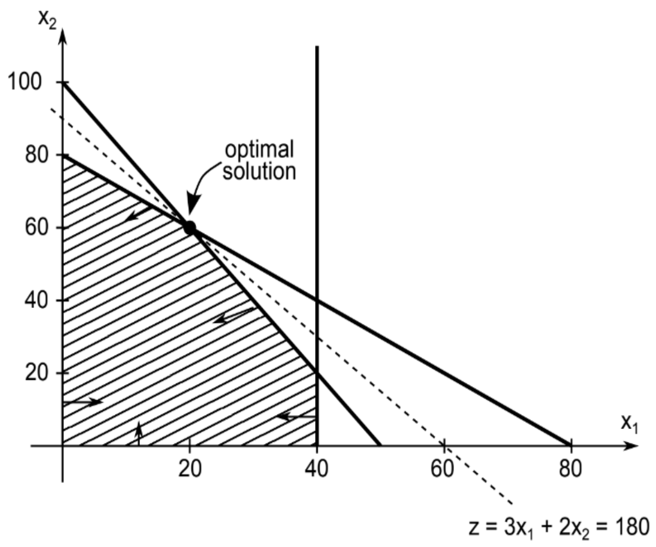
- Consider this LP:

$$\max z = 3x_1 + 2x_2$$
$$\text{s.t.} \quad 2x_1 + x_2 \leq 100$$
$$x_1 + x_2 \leq 80$$
$$x_1 \leq 40$$
$$x_1, x_2 \geq 0$$



? Can you find the range for x_1's coefficient in the objective function (c_1) where the current solution will stay the same?

Range of Optimality For c_1



max $z = 3x_1 + 2x_2$

s.t. $2x_1 + x_2 \leq 100$

$x_1 + x_2 \leq 80$

$x_1 \leq 40$

$x_1, x_2 \geq 0$

- The slope of the objective function line is $-\dfrac{c_1}{c_2}$
- The slope of the 1st binding constraint, $x_1 + x_2 = 80$, is -1.
- The slope of the 2nd binding constraint, $2x_1 + x_2 = 100$, is -2.

Range Of Optimality For c_1

- The slope of the objective function line is $-\frac{c_1}{c}$
- The slope of the 1st binding constraint, $x_1 + x_2^2 = 80$, is -1.
- The slope of the 2nd binding constraint, $2x_1 + x_2 = 100$, is -2.

- Find the range of values for c_1 (with c_2 staying 2) such that the objective function line slope lies between that of the two binding constraints:
  $-2 \leq -\frac{c_1}{2} \leq -1$

- Multiplying through by -2 (and reversing the inequalities):
  $2 \leq c_1 \leq 4$

Range of Optimality For c_1

- Find the range of values for c_2 (with c_1 staying 3) such that the objective function line slope lies between that of the two binding constraints:

$$-2 \leq -\frac{3}{c_2} \leq -1$$

Range of Optimality For c_1

- Find the range of values for c_2 (with c_1 staying 3) such that the objective function line slope lies between that of the two binding constraints:

$$-2 \leq -\frac{3}{c_2} \leq -1$$

- Multiplying by -1: $2 \geq \frac{3}{c_2} \geq 1$
- Inverting, $\frac{1}{2} \leq \frac{c_2}{3} \leq 1$
- Multiplying by 3: $\frac{3}{2} \leq c_2 \leq 3$

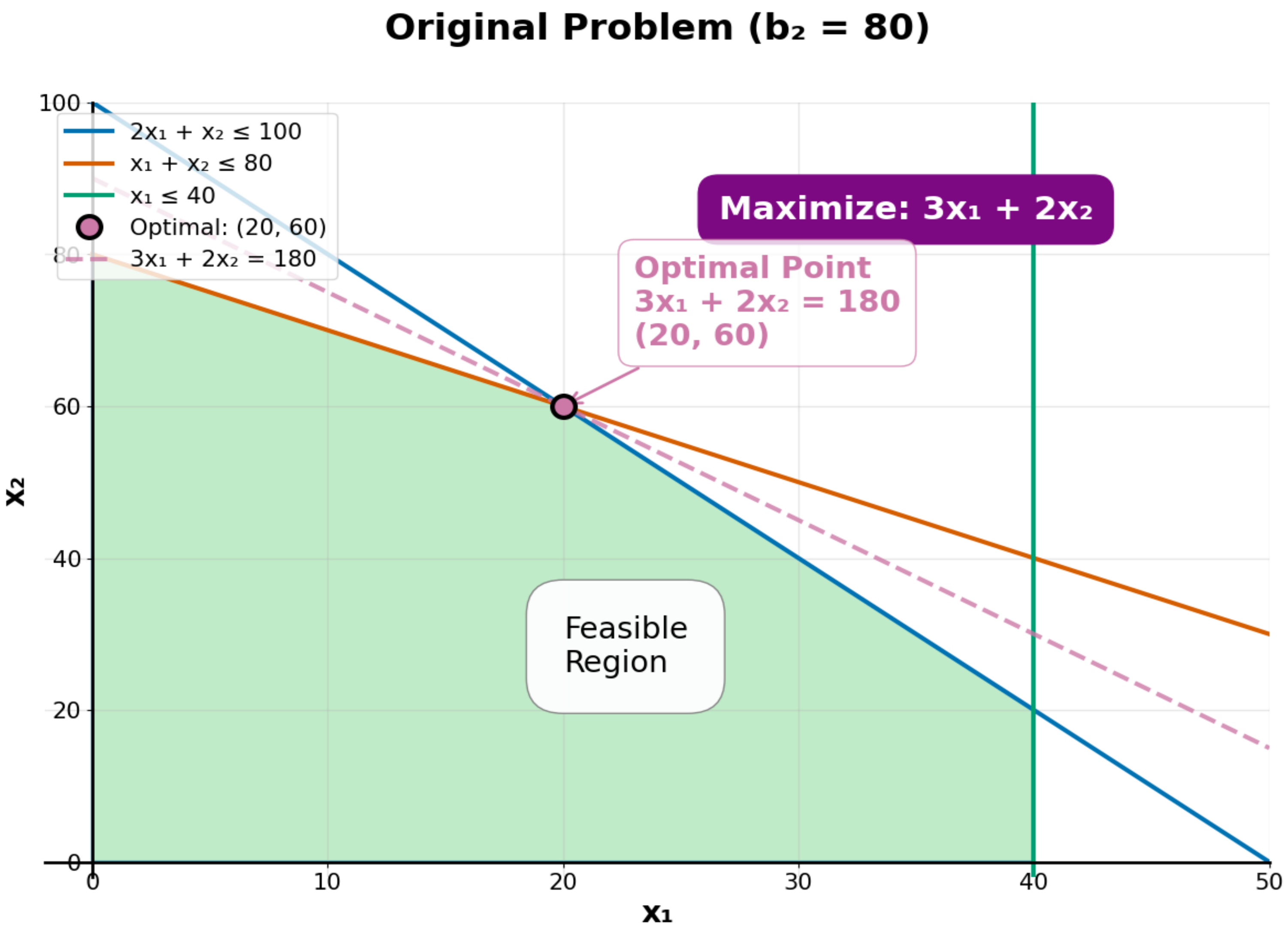
Changing The RHS

- Consider this LP:

$$
\begin{aligned}
\max z &= 3x_1 + 2x_2 \\
\text{s.t.} \quad & 2x_1 + x_2 \le 100 \\
& x_1 + x_2 \le 80 \\
& x_1 \le 40 \\
& x_1, x_2 \ge 0
\end{aligned}
$$

Let's change the b_2 on the RHS of second constraint

Changing The RHS

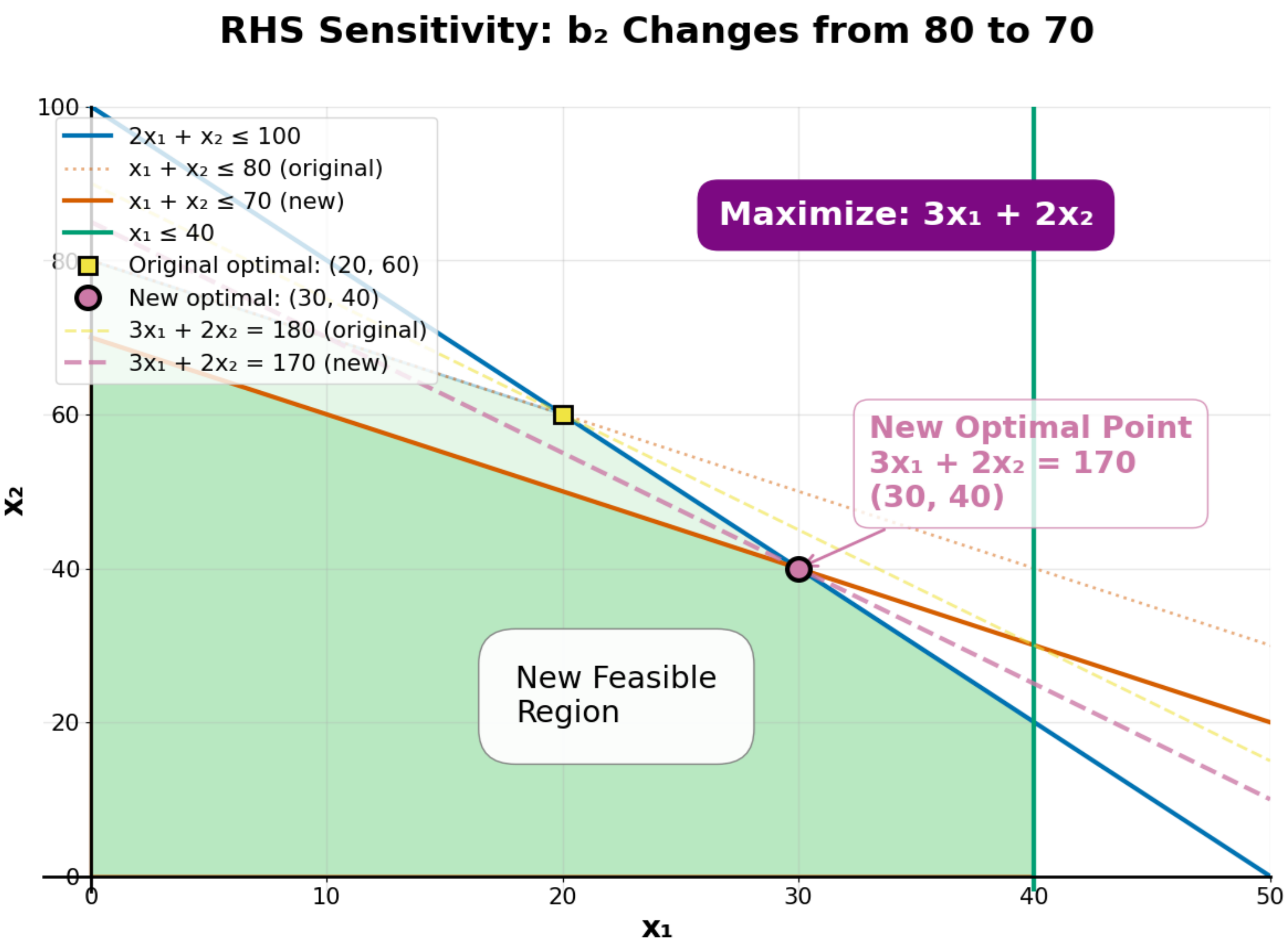


Changing The RHS

$$\begin{aligned} \max\ z &= 3x_1 + 2x_2 \\ \text{s.t.}\quad 2x_1 + x_2 &\le 100 \\ x_1 + x_2 &\le 80 \\ x_1 \quad\quad &\le 40 \\ x_1, x_2 &\ge 0 \end{aligned}$$

Change the coefficient b_2 = 80 on the RHS of the 2nd constraint to 70: $x_1 + x_2 \le 70$

- the point $x_1 = 20$, $x_2 = 60$ **not feasible**
- new optimum $x_1 = 30$, $x_2 = 40$ but **same basis** $\{x_1, x_2, x_5\}$



Changing The RHS

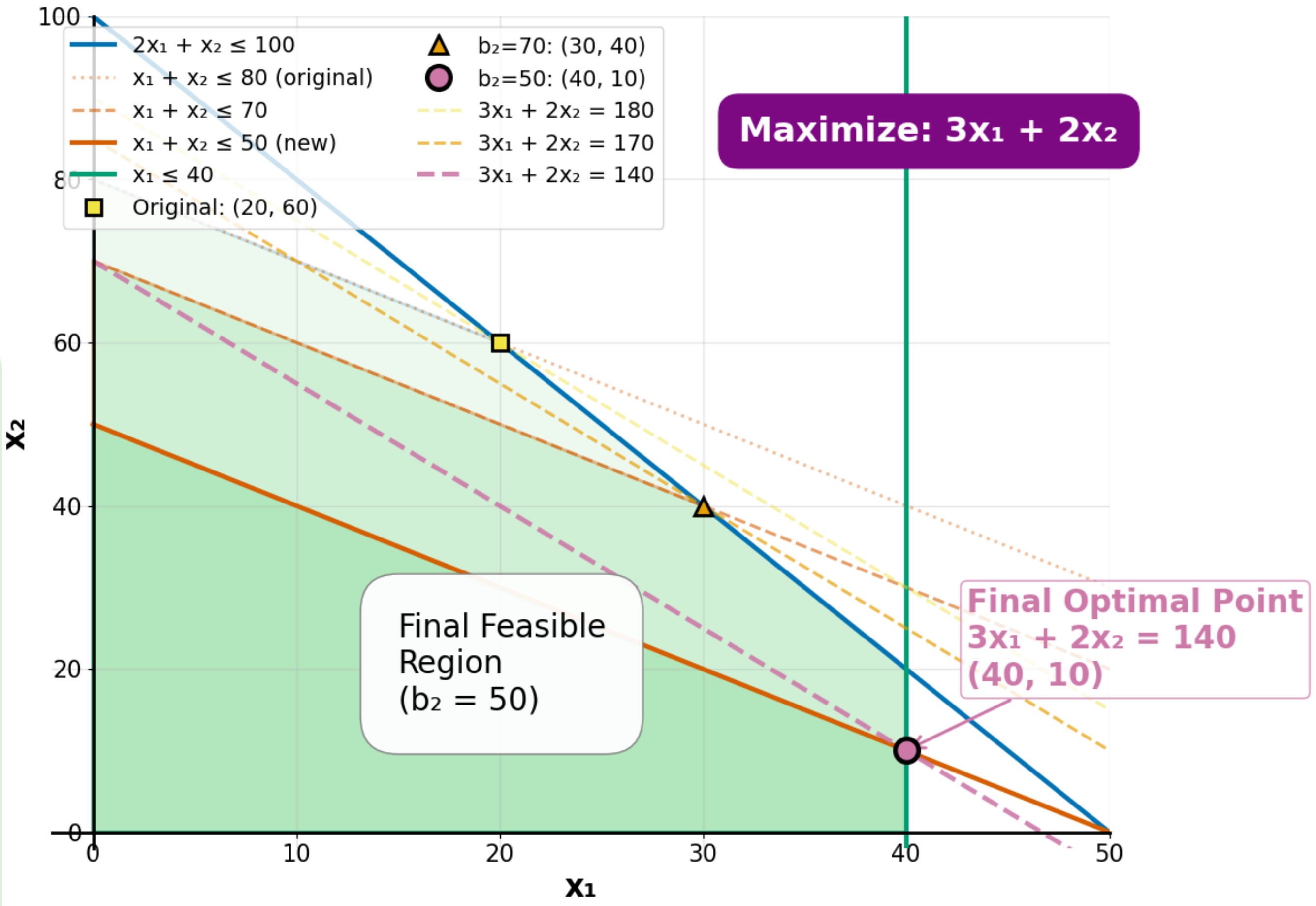
$$\max z = 3x_1 + 2x_2$$

$$\text{s.t.} \quad \begin{aligned} 2x_1 + x_2 &\le 100 \\ x_1 + x_2 &\le 80 \\ x_1 \quad &\le 40 \\ x_1, x_2 &\ge 0 \end{aligned}$$

Change the coefficient b_2 = 50: $x_1 + x_2 \le 50$

- the point x_1 = 20, x_2 = 60 **not feasible**
- new optimum x_1 = 40, x_2 = 10 **new basis** $\{x_1, x_2, x_3\}$

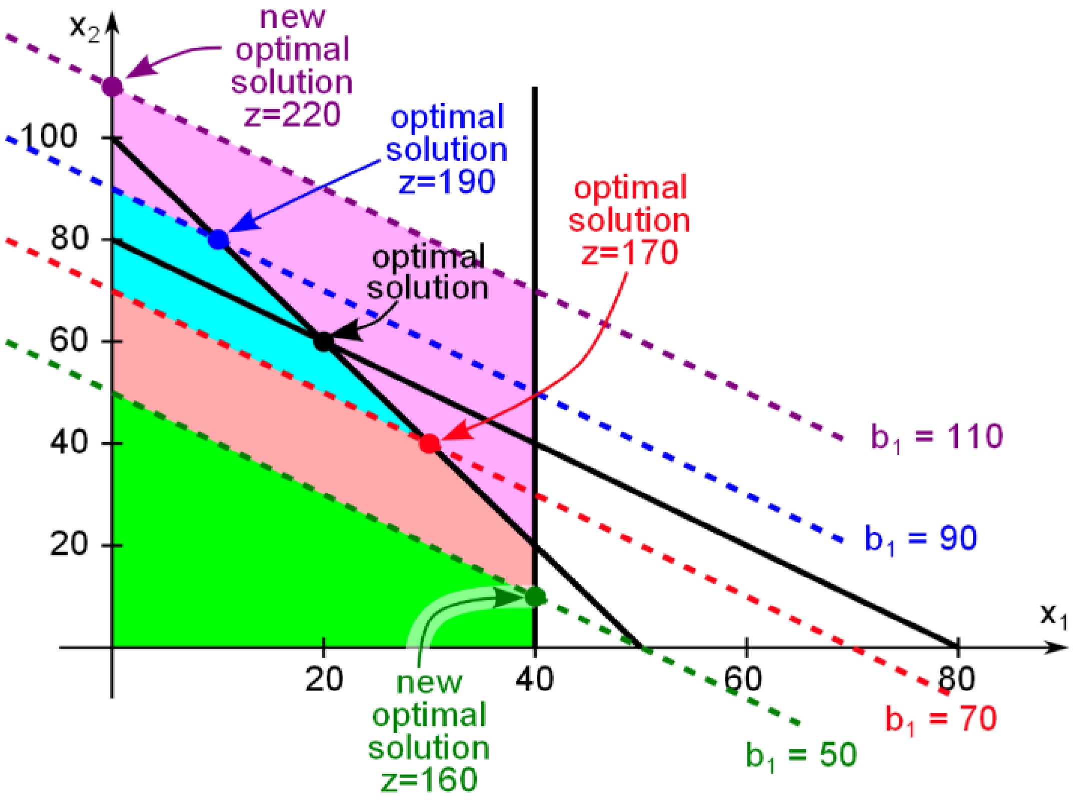
RHS Sensitivity: b₂ Changes from 70 to 50: b₂ = 80 → 70 → 50



Changing The RHS

- Consider this LP:

$$\max z = 3x_1 + 2x_2$$
$$\text{s.t.} \quad 2x_1 + x_2 \le 100$$
$$x_1 + x_2 \le 80$$
$$x_1 \le 40$$
$$x_1, x_2 \ge 0$$



Change the coefficient b_2 = 90: $x_1 + x_2 \le 90$

- the point $x_1 = 20, x_2 = 60$ **feasible**
- new optimum $x_1 = 10, x_2 = 80$ **same basis** $\{x_1, x_2, x_5\}$

Sensitivity Analysis Using Solvers

Example | Portfolio Optimization

Let’s visit an example:

Suppose you are a financial engineer advising a client on asset allocation. Your client has $100,000 to invest and you have narrowed the options down to three different investment vehicles: stocks, bonds, and real estate.

The expected annual returns for these investment options are as follows:

- Stocks: 8% annual return
- Bonds: 5% annual return
- Real estate: 6% annual return

Example | Portfolio Optimization

The client has set the following constraints:

1. At least 30% of their portfolio should be in bonds to maintain a conservative posture.
2. No more than 50% of the portfolio should be in stocks to minimize risk.
3. At least 10% should be invested in real estate for diversification.

Example | Portfolio Optimization

Mathematical Model

Let x_1 , x_2 , and x_3 represent the proportions of the portfolio to be invested in stocks, bonds, and real estate, respectively.

Objective Function:

Maximize the expected return of the portfolio while adhering to the client's constraints.

Maximize: $Z = 0.08x_1 + 0.05x_2 + 0.06x_3$

Example | Portfolio Optimization

Subject to Constraints:

$$
\begin{aligned}
x_1 + x_2 + x_3 &= 1 && \text{(The sum of the proportions must be 1)} \\
x_2 &\geq 0.3 && \text{(At least 30\% in bonds)} \\
x_1 &\leq 0.5 && \text{(At most 50\% in stocks)} \\
x_3 &\geq 0.1 && \text{(At least 10\% in real estate)} \\
x_1, x_2, x_3 &\geq 0 && \text{(Proportions cannot be negative)}
\end{aligned}
$$

Example | Portfolio Optimization

The full LP formulation is given below. Let's switch to Gurobi to see how to set up the sensitivity analysis.

Maximize $Z = 0.08x_1 + 0.05x_2 + 0.06x_3$

Subject to

$$
\begin{aligned}
x_1 + x_2 + x_3 &= 1 && \text{(The sum of the proportions must be 1)} \\
x_2 &\geq 0.3 && \text{(At least 30\% in bonds)} \\
x_1 &\leq 0.5 && \text{(At most 50\% in stocks)} \\
x_3 &\geq 0.1 && \text{(At least 10\% in real estate)} \\
x_1, x_2, x_3 &\geq 0 && \text{(Proportions cannot be negative)}
\end{aligned}
$$

Applying Monte Carlo to LPs

Monte Carlo Simulation

This type of simulation is a model used to predict the probability of a variety of outcomes when the potential for random variables is present

It is a computational technique that runs multiple random trials or iterations

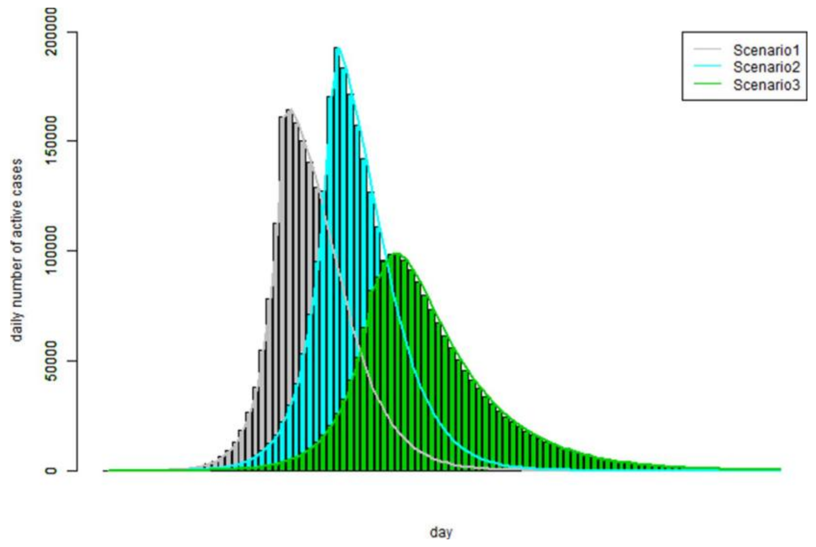
Some fields that use MCS are risk analysis, finance, and physics, among others

SCIENTIFIC REPORTS

natureresearch

A novel Monte Carlo simulation procedure for modelling COVID-19 spread over time

Link



Example | Essential Goods Procurement for A Shelter

- Let’s revisit the shelter problem from Assignment 1 and assume that the cost is not a known parameter, but a random parameter that’s uniformly distributed.